A-A-2348
February 15, 1985

COMMERCIAL ITEM DESCRIPTION

INDICATOR, DIAL, ACCESSORIES, AND TEST SETS

The General Services Administration has authorized the use of this commercial item description as a replacement for Military Specification MIL-I-18422 which is canceled.

This commercial item description covers dial indicators, accessories, and test sets intended for use in aligning work pieces on machine tools, inspecting fixtures, and checking dimensions and variations from a reference point.

Salient characteristics.

Dial indicators.

Dial indicators shall comply with the requirements of ANSI B89.1.10, issue in effect on the date of invitation for bid or request for proposals, with the following additional requirements, and shall be furnished in the following types and styles as specified:

Type I - 0.0001 inch graduations
Type II - 0.0005 inch graduations
Type III - 0.001 inch graduations
Type V - 0.00005 inch graduations

Style A. Spindle parallel to dial. Style A indicators shall be supplied with a regular back (central vertical bracket).

Style B. Spindle perpendicular to dial from back of case. The holding rod shall be 1/4 inch diameter by 1-1/2 inch long, minimum, as measured from the bezel edge to the end of the rod. When specified, the item shall accommodate hole measuring accessories.

Style C. Universal, adjustable contact point (test indicator). The contact point shall hold any position within a 220 degree arc. Style C indicators shall be quick acting, manual reversing types. The readings shall be unaffected by magnetic reading (nonmagnetic type) as specified.

Dial indicators supplied in accordance with this description shall be so constructed as to preclude the excessive entry of dust and dirt into the mechanism. Jeweled bearings, cushioned (antishock) movement, and plunger contact point action shall be supplied as specified. Crystals shall be clear, transparent and unbreakable. Bezels shall have knurls or serrations around the periphery.

Dial indicators shall comply with the requirements of Tables I through III.

TABLE I

Type II, Style A.

National Stock Number (NSN)	Size Group	Dial Reading	Spindle travel (inch)	Maximum force to deflect spindle (grams)	
				25%	100%
5210-00-293-3016	2	0-25-0	0.125	105	120

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Salient characteristics - continued

Type III, Style A.

National Stock Number (NSN)	Size Group	Dial Reading	Spindle travel (inch)	Maximum force to deflect spindle (grams)	
				25%	100%
5210-00-243-9649	2	0-100	1.000	115	260

TABLE II

Type III, Style B

National Stock Number (NSN)	Bezel dia.	Dial Reading	Spindle travel (inch)	Maximum force to deflect spindle (grams)	
				25%	100%
5210-00-277-8840	1-3/8 to 2-3/8	0-100	0.200	125	190
5210-00-377-6525	1-3/8 to 2-3/8	0-50-0	0.200	125	190

TABLE III

Style C.

National Stock Number (NSN)	Type	Bezel dia. (inch)	Dial Reading	Lever travel min. (inch)	Maximum force to deflect lever upward (grams)	
					25%	100%
5210-00-273-9791	I	7/8 to 1-3/8	0-4-0	0.008	18	23
5210-00-223-9648	III	7/8 to 1-3/8	0-15-0	0.030	22	30

Testing. Dial indicators shall be tested for accuracy, repeatability, and the force required to deflect the spindle or lever using the following materials and procedures.

Materials.

Gage blocks - Range: 0.10 to 4.0 inches
4.0 to 20.0 inches
Accuracy: ± 0.000006 inch

Surface plate - Dimensions: 24 X 36 inches
Accuracy: ±0.000025 inch

Vernier height gage - Range: 0 to 18 inches
Accuracy: ±0.001 inches/10 inches

Mounting hardware - Suitable to mount all types of indicators to the height gage.

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Procedures.

Accuracy.

Mount the dial indicator on the height gage or other suitable holding device using hardware that will result in a rigid and stable set-up. Place the set-up on the surface plate. Wring together a suitable combination of gage blocks to provide a calibration starting point. Adjust the set-up so the lever or spindle makes contact with the gage blocks. Using the set-up, adjust the indicator such that the pointer is deflected a distance equal to one-half the difference between the specified range and the actual range of the indicator. Lock the set-up in this position. Adjust the indicator's bezel until the pointer reads "zero" on the dial.

Add the appropriate combinations of gage blocks to the set-up to provide 8 evenly spaced check points over the full range of the indicator. Readings observed at each check point shall be within the limits as specified below.

Without resetting the indicator, remove the gage blocks in a way that will provide readings at the 8 check points established above as the spindle is deflected outwards. Accuracy shall be as specified below plus an additional 1/3 graduation to compensate for repeatability tolerances.

Repeatability.

Mount the indicator as described above. Using gage blocks, deflect the indicator spindle through approximately 25, 50 and 75 percent of the full range. Repeat each deflection 5 consecutive times at each check point. Readings shall be within the limits as specified below.

Spindle or lever load test. Using the set-up described above, position the indicator such that the spindle or lever makes contact with one platform of an analytical balance. Load the opposite platform with weight sufficient to deflect the spindle or lever the distances specified for each NSN in Tables I through III. To be considered acceptable, the weight required to deflect the spindle or lever the specified distances shall be less than the weight limits specified for each NSN in Tables I through III.

Requirements. Dial indicators shall comply with the following requirements for accuracy and repeatability when tested as specified above for a minimum of 500,000 deflections.

Accuracy requirements. All types and styles of dial indicators shall not be in error by more than plus or minus one dial graduation throughout the entire range of the indicator. However, indicators having a range exceeding 2-1/3 revolutions (long range indicators), shall not be in error by more than plus or minus one dial graduation for the first 2-1/3 revolutions and not more than 5 dial graduations for travel in excess of 2-1/3 revolutions.

Repeatability requirements. Readings at any point within the entire range of the indicator shall be reproducible through successive spindle deflections to within a plus or minus 1/5 dial graduation.

Accessories. The following accessories shall be furnished with the dial indicators as specified in Table IV.

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Accessories - Continued

Contact points. Unless otherwise specified, standard 1/4 inch long contact points shall be supplied with Style A and B indicators. When specified, button, cone or ball contact points shall be supplied. Extra length contact points for Style C indicator shall be supplied when specified.

Hole attachment. When specified, a hole measuring attachment with a depth capacity of 1-11/16 ±1/16 inch shall be supplied with Style B indicators. The device shall be capable of being securely attached to the spindle housing.

Stanchion bases. When specified, a stanchion base shall be supplied with the dial indicator and shall consist of the following components:

A base and upright column 0.465 inches in diameter by 8 inches long minimum with provision for attaching the column securely to the base.
A dial holding rod 3/8 inch diameter by 8-1/4 inches long minimum with a thumbscrew clamp for attaching the indicator and a sliding swivel clamp for holding the indicator rod to the upright column in any fixed position by tightening a thumbscrew or screws. The clamp and dial holding rod shall be capable of turning and swiveling 360 degrees on the upright column. The assembly shall contain a reducing bushing as specified.

The standard stanchion base shall be 2-3/16 inches by 8-1/4 inches minimum with a longitudinal slot top and bottom for attaching the upright column.

Arbor clamp. When specified, an arbor clamp shall be furnished with the indicator. The clamp shall be essentially a "C" clamp with a 5/16 inch diameter by 5 inches long minimum clamping rod. A "V" notch shall be opposite the clamping rod. The capacity of the clamp shall be not less than 1-1/4 inches.

Tool post. Tool post mounting for Style C indicators shall be supplied when specified. The tool post mounting shall be either a holding bar with swivel clamp or universal friction holder with a universal swiveling arm and body clamp as specified.

Test sets (tool post type), Style B indicators. Style B indicator test sets shall be supplied with the following components:

(1) Style B indicator as specified.
(1) 1/4 inch by 5/16 inch clamp (complete sleeve)
(1) Hole attachment
(1) Arbor clamp
(1) Tool post holder, 3/8 inch by 3/4 inch by 5-3/4 inches long minimum, with 5/16 inch diameter by 4 inches long minimum, upright post.

Cases. When specified, cases shall be supplied with the indicators or test sets. The case shall possess sufficient strength and durability to withstand the conditions encountered during normal shop use. It shall be designed to house compactly, and hold securely, all component parts, as specified, including the indicator.

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TABLE IV

National Stock Number (NSN)	Type. Style. Accessories required
5210-00-293-3016	Type II, Style A, 5/32 inch long regular shape contact point, standard stanchion base without base stops, with offset arm, reducing bushing, tool post holder, jeweled bearings.
5210-00-243-9649	Type III, Style A, with revolution counter.
5210-00-277-8840 (Test set)	Type III, Style B, plunger type contact point action, standard, button and ball contact points, hole attachment, arbor clamp, tool post holder with upright post, clamp, jeweled bearings, case.
5210-00-377-6525	Type III, Style B, same accessories as 5210-00-277-8840 but with cone contact point in lieu of ball contact point, plunger type contact point action, jeweled bearings.
5210-00-273-9791	Type I, Class C, nonmagnetic type with ball contact point, case, lever gear assembly and hairspring made of non-magnetic material, and with holding bar with swivel clamp, cushioned movement, jeweled bearings.
5210-00-223-9648	Type III, Style C, ball contact point, universal friction tool post holder with universal swivel arm with clamping attachment.

Workmanship. Dial indicators, accessories, and test sets shall be free from any defect that impairs durability, serviceability, and appearance. All moving parts shall operate freely and without restriction.

Marking. When size permits, each indicator and accessory shall be marked in a plain, permanent and legible fashion with the manufacturer's name or trademark. The country of origin shall be marked on each dial indicator.

Regulatory requirement. The contractor/supplier is encouraged to utilize recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Packaging, packing, labeling and marking. Packaging, packing, labeling and marking shall be as specified in the contract or order.

Custodian

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Review Activities

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PREPARING ACTIVITY:

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